

Problem

Find the Laplace transform of the signal in Fig. 15.26.

Figure 15.26:



Step-by-step solution

Step 1 of 2

Refer to Figure 15.26 in the text book.

The expression of $f(t)$ is,

$$\begin{aligned} f(t) &= \left(\left(\frac{25}{2} - 0 \right) tu(t) + \left(-\frac{25}{4} - \frac{25}{2} \right) (t-2)u(t-2) \right) \\ &\quad + \left(0 - \left(-\frac{25}{4} \right) \right) (t-6)u(t-6) \\ &= 12.5tu(t) - 18.75(t-2)u(t-2) + 6.75(t-6)u(t-6) \end{aligned}$$

Step 2 of 2

Apply Laplace transform on both sides.

$$\begin{aligned} F(s) &= 12.5 \frac{1}{s^2} - 18.75e^{-2s} \frac{1}{s^2} + 6.75e^{-6s} \frac{1}{s^2} \\ &= \frac{12.5 - 18.75e^{-2s} + 6.75e^{-6s}}{s^2} \end{aligned}$$

Thus, the expression of the Laplace transform of the signal is

$\frac{12.5 - 18.75e^{-2s} + 6.75e^{-6s}}{s^2}$
